

602 Project

602 Project Proposal

Research Question

I will focus on the following questions: - Which agencies have the highest average salaries? - Which agencies spend the most on employee compensation? - Among selected data science-related civil service titles, which agencies offer the highest compensation?

Selected Civil Service Titles

- Data Analyst
- City Research Scientist
- Research Scientist
- Statistician
- IT Project Specialist

Justification

I work for the City of New York, so I'm interested in the salary ranges between the different agencies.

Data Sources

Using the following [NYC Opendata source](#), which is an open datasource created by NYC's Office of Payroll Administration (OPA)

Libraries

- pandas
- numpy
- matplotlib (maybe seaborn)

EDA & Summary Statistics

Raw Dataset Schema

Field Name	Type
fiscal_year	numeric
payroll_number	numeric
agency_name	string
last_name	string
first_name	string
mid_init	string
agency_start_date	datetime
work_location_borough	string
title_description	string
leave_status_as_of_june_30	string
base_salary	numeric
pay_basis	string
regular_hours	numeric
regular_gross_paid	numeric
ot_hours	numeric
total_ot_paid	numeric
total_other_pay	numeric

Analysis Dataset Schema

I will reduce the dataset to the following fields: - fiscal_year → **year** - agency_name → **agency**
- title_description → **title**
- base_salary → **salary**
- regular_gross_paid → **salary_paid**

I will also create derived variables: - employee_id - index - other_paid
- total_ot_paid + total_other_pay
- total_paid
- salary_paid + other_paid

Analysis

- Grouped summary tables by `agency_name` and `title_description` including:
 - count of `employee_id`
 - mean `base_salary`
 - median `base_salary`
 - minimum `base_salary`
 - maximum `base_salary`
 - standard deviation of `base_salary`
 - sum of `total_paid`
- Ranked summary tables derived from the grouped summaries, including:
 - Top 5 agencies by mean `base_salary`
 - Top 5 agencies by sum of `total_paid`
 - Top 5 agencies within the selected `title_description` values by mean `base_salary`
 - Top 5 agencies within the selected `title_description` values by mean `total_paid`
- Distribution visualizations, such as:
 - bar charts for top 5 agency rankings
 - boxplots of salary distributions by agency
 - histograms showing salary distributions overall or within selected titles

Presentation + Submission

For the presentation, I will render the Quarto document as an HTML slide deck using reveal.js. An example of this format can be found [here](#).

For the final submission, I will provide a PDF generated from a Quarto (.qmd) document with inline Python code and visualizations.